

## Highlights

### **Your Perception Monitor May Be Measuring Nothing: A Null-Detector Diagnostic and a Cross-Session Validity Failure in Conformal Monitoring for Autonomous Navigation**

Author names withheld pending anonymity-policy confirmation

- A null-detector diagnostic exposes silent failure in conformal perception monitors
- Zeroing the detector reproduces monitor output bit-for-bit under score collapse
- Cross-session exchangeability failure inflates miscoverage under pooled calibration
- Session-conditional calibration gives inconsistent gains at this sample size
- A drive-disjoint weather-effect estimate is an underpowered null, not an absence

# Your Perception Monitor May Be Measuring Nothing: A Null-Detector Diagnostic and a Cross-Session Validity Failure in Conformal Monitoring for Autonomous Navigation

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## Abstract

Autonomous navigation stacks increasingly monitor perception with conformal prediction: calibrate a nonconformity score, track its miscoverage with an anytime-valid sequential test, and alarm when perception becomes untrustworthy. We set out to reduce detection delay for such a monitor on camera/LiDAR 3D detection under real adverse weather; we did not, and report two silent failure modes instead. First, when a detector’s regression head degenerates, its nonconformity score reduces to a function of ground-truth labels alone, and every downstream component – calibration, sequential betting, alarm timing – behaves exactly as specified while measuring nothing about perception. We introduce a null-detector diagnostic, a one-pass control that reruns the pipeline with the detector’s output zeroed, and show it reproduces our own earlier results bit-for-bit. Second, split conformal’s exchangeability assumption fails across driving sessions on the Canadian Adverse Driving Conditions dataset: per-drive miscoverage under one pooled quantile varies several-fold across drives that are all nominal, with between-session variation comparable to the weather effect sought, so the guarantee remains formally correct while becoming operationally useless – the monitor tracks route identity, not perception loss. On the weather effect itself we report an underpowered null, not a demonstrated absence, and show why an earlier, in-sample version of the same measurement had appeared significant. We recommend both the null-detector diagnostic

and session-conditional evaluation as standard practice for conformal perception monitoring in navigation.

*Keywords:* Conformal prediction, Perception monitoring, Autonomous navigation, Exchangeability, Anytime-valid sequential testing, Adverse weather

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## 1. Introduction

An autonomous navigation stack is architected to trust its perception layer. Path planning, risk assessment and minimum-risk manoeuvre selection all consume detections as though they were reliable, and the safety case for the vehicle rests on that reliability holding under the conditions the vehicle actually meets. Because it will not always hold, a monitoring layer is attractive: something that watches perception during operation and tells the planner when its inputs have stopped being trustworthy, so the stack can slow down, hand back, or pull over.

Conformal prediction offers a principled way to build one. Calibrate a nonconformity score on nominal data, and the fraction of new predictions falling outside the conformal region is controlled at a chosen level without distributional assumptions. Pair it with anytime-valid sequential testing — a betting process whose wealth is a nonnegative supermartingale under the nominal hypothesis — and the monitor can be interrogated continuously without a multiple-testing penalty, which is exactly what a vehicle needs from a system asked “has anything changed yet?” at every frame.

We set out to reduce the detection delay of such a monitor for camera/LiDAR 3D detection under real adverse weather. We did not achieve that, and this paper is about what we found instead. Two findings concern how monitors of this design fail *silently* — continuing to emit calibrated quantiles and plausible alarm times while not doing their job — and both matter more for navigation than a delay improvement would have.

*A monitor can measure nothing and look healthy.* The nonconformity score used throughout this literature is a residual between predicted and ground-truth boxes. If the detector’s regression head degenerates — as ours did, under

a placeholder training loss whose optimum is a constant — that residual becomes  $\|B_{\text{target}}\|_1$ , a function of the labels alone. Every downstream component then behaves exactly as specified: calibration returns a well-defined quantile, because it is the quantile of *some* exchangeable score distribution; the betting process  
 30 accumulates wealth, because the miscoverage rate still moves; the alarm fires at a plausible time, because annotation statistics genuinely differ between a dataset’s nominal and adverse splits. The monitor reports dataset structure as perception degradation, and nothing in its output reveals this.

We introduce a *null-detector diagnostic* that does reveal it: rerun the whole  
 35 pipeline, calibration included, with the prediction head’s output replaced by zeros. It costs one evaluation pass and needs no extra data, training or annotation. Applied to our own earlier results it reproduced them bit-identically — a nominal-to-degraded jump of 0.13904668284371138 from both the real and the zeroed detector, agreeing to 17 significant figures (Section 4.2). A monitor  
 40 with no detector performed identically to one with a detector, and the reason is instructive rather than mysterious: content that is homogeneous within a category produces a score that looks stable across sessions in a way genuine, session-heterogeneous detector error does not, so a null detector’s output can in principle look *no worse behaved* than the real pipeline’s even though it carries  
 45 no perceptual information at all (Section 5.1) — precisely why the failure is invisible from output inspection alone.

*Exchangeability fails across driving sessions..* Conformal validity is conditional on exchangeability between calibration and test data. Real driving data arrives in sessions — drives — that differ persistently in route, traffic, lighting and  
 50 weather. On CADC [1], on a drive-disjoint held-out split, we measure per-drive miscoverage under a single pooled quantile spreading 3–5× across drives that are all nominal, all clear-road, on a nominal-drive sample thin enough (four held-out drives) that we report the range as indicative rather than precise (Section 4.3). Between-session variation is comparable in size to the weather effect the monitor  
 55 exists to detect. The guarantee stays formally correct — it holds on average

over the calibration distribution — and becomes operationally useless, because the monitored quantity is dominated by which session the vehicle is on.

For navigation specifically this is the worst way for a monitor to fail. An alarm that tracks route identity rather than perception loss will fire on an unremarkable drive and stay silent on a dangerous one, and a planner wired to  
60 react to it inherits that error directly. Worse, a monitor with undemonstrated sensitivity that is nonetheless counted in a safety argument is more dangerous than no monitor at all, because the false assurance displaces other mitigations.

*What we can and cannot say about the weather effect..* Our original objec-  
65 tive requires a detectable difference between nominal and degraded conditions. On drive-disjoint held-out data, with resampling clustered over both nominal sessions and degraded drives, we measure  $+0.069$  with a 95% interval of  $[-0.073, +0.210]$ . This is an *underpowered null*, not a demonstrated absence, and we are careful to report it as such throughout: the interval is consistent  
70 with a real effect of the size we previously claimed and equally consistent with none. Nine evaluation sessions cannot resolve an effect of this magnitude — the minimum detectable effect at 80% power, computed directly from this interval’s bootstrap standard error, is roughly 0.20, almost three times the point estimate.

We also report why an earlier version of this measurement appeared signif-  
75 icant. Two defects compounded: the detector had trained on the very drives being evaluated, and a bookkeeping error in the bootstrap let each session’s own calibration frames re-enter the window being monitored. Reporting the interval both with and without the degraded arm resampled shows that even the narrower comparison fails to reach significance once the drive-disjoint split  
80 is applied (Section 4.5). We suggest dual reporting as a general practice for clustered effect estimates: it helps a reader distinguish an underpowered design from a genuine absence, which the point estimate alone never does.

*Scope and honesty about this work’s own history..* Both findings above were discovered by controls applied to our own pipeline, and the numbers those con-  
85 trols invalidated were ours. We think this strengthens rather than weakens the

case for the controls, but it puts an obligation on us to be exact about what is being corrected: the affected results were obtained in an unpublished predecessor of this work and were never published, so no external record requires correction. Snowy Scenes [2] appears throughout as a negative control: its null  
90 detector shows a large, significant content-driven effect where its real detector shows none, demonstrating that the diagnostic fires on a genuine artifact when one is present. CADC’s corrected result is more modest than we first reported it to be — neither its real nor its null detector reaches significance — which at least shows the diagnostic does not fire spuriously when the honest answer is  
95 that we cannot tell.

## 2. Related Work

### 2.1. Robust Multimodal Fusion for Autonomous Perception

Camera/LiDAR fusion for 3D detection has moved from simple feature concatenation toward BEV-centric architectures that project both modalities into  
100 a shared spatial representation before fusing, with attention-based mechanisms used to weight each modality’s contribution per region. A separate line of work has shown that such fusion is not automatically robust: a corrupted or adversarially perturbed single modality can dominate a static or softly-attended fusion decision and collapse detection performance, since nothing in the fusion mechanism  
105 verifies that the modalities being combined actually agree geometrically before combining them. The detector this paper monitors adopts a verify-then-fuse design intended to address exactly that failure mode – a Cross-modal Consistency Probe that computes a per-BEV-cell consistency score between camera and LiDAR features, feeding a Gated Adaptive Fusion Module that quarantines  
110 a disagreeing modality at the grid-cell level, trained under an Adversarial Consistency Training objective. We treat this design here strictly as background infrastructure and describe only as much of its mechanism (Section 3) as this paper’s own monitoring layer depends on, since the monitor itself is detector-agnostic.

115 *2.2. Conformal Prediction and Anytime-Valid Sequential Testing*

Split conformal prediction [3, 4] gives distribution-free, finite-sample coverage guarantees for a black-box predictor’s outputs by calibrating a nonconformity score on held-out exchangeable data; the guarantee is a batch one, valid at a single fixed evaluation, and does not by itself say anything about monitoring a  
120 live, continuously arriving stream. Sequential testing by betting [5, 6] closes that gap: framing a hypothesis test as an accumulating wealth process that is a non-negative supermartingale under the null lets Ville’s inequality bound the false-alarm probability uniformly over all stopping times, giving an anytime-valid guarantee with no fixed monitoring horizon and no post-hoc multiple-testing  
125 correction. Closely related, WATCH [7] adapts weighted-conformal martingales to monitor deployed AI systems for distribution shift, reweighting the martingale under a known or estimated covariate-shift model rather than betting directly on a miscoverage signal; it targets general AI deployment monitoring rather than perception specifically, and does not address either of this paper’s  
130 two central failure modes (a degenerate score masquerading as signal, or exchangeability breaking across structured sessions). The direct predecessor for applying this backbone to a live perception-failure problem is Monroy Muñoz, Verma, and Timans’s work on sequential object-tracking-failure detection [8], which frames tracking failure as an e-process over a bounded tracking-quality  
135 metric with the betting rate learned adaptively from the metric’s own history (approximate GRAPA or Scale-Free Online Gradient Descent), evaluated across four 2D visual-tracking benchmarks. It is a strong, directly applicable baseline, and everywhere in this paper that we compare against a "covariate-blind" betting rule, we mean their rule, reimplemented exactly, not a strawman.

140 Adjacent recent work conditions conformal guarantees on covariates or system state without adopting the sequential/anytime-valid framing: state-dependent conformal prediction for perception-based autonomous systems [9] combines conformal error bounds with symbolic reachability analysis but remains a batch procedure; context-aware nonconformity functions for robotic planning and perception  
145 [10] learn context-conditioned nonconformity scores, with context drawn

from learned task representations rather than a cross-modal sensor-disagreement signal specifically; and a value-gated modality-refiner approach [11] uses cross-modal agreement to gate fusion, but as a learned neural gate rather than a statistical test covariate, and targets sentiment/action-recognition/audio-visual tasks rather than autonomous-vehicle perception. None of these combine sequential anytime-valid testing, spatially-resolved multi-object structure with multiplicity control, and a betting rule conditioned on an external cross-modal disagreement signal – the combination this paper’s monitoring layer targets, following the sharpened contribution statement worked out in the ICRA-targeted predecessor to this paper’s monitoring contribution.

### 2.3. Conditional Validity and Negative Controls

Marginal conformal validity holds on average over the calibration distribution and can fail badly within identifiable subgroups. Mondrian conformal prediction [12] addresses this by conditioning calibration on a taxonomy, giving validity within each category rather than marginally; the textbook treatment appears in Vovk, Gammerman and Shafer [3]. The taxonomy is usually a class label. We apply it with the driving session as the category, which is the grouping along which we measure exchangeability to fail in real autonomous-driving data (Section 4.3), and we report a case where the resulting benefit reverses.

We are not aware of prior work that reports a null-detector control for a conformal perception monitor – that is, rerunning the pipeline with the monitored model’s predictions replaced by a constant to establish that the monitor responds to model error rather than to dataset structure. The closest established practice is the use of permutation and label-shuffling baselines in the broader evaluation literature, which serve the same epistemic purpose: distinguishing a method’s measured effect from structure already present in the data. Our contribution here is not the idea of a negative control but the specific, cheap instantiation for conformal monitoring, together with evidence that it changes conclusions on real published-style numbers (Section 4.2).



175 *2.4. Adverse-Weather Datasets for Autonomous Perception*

Large-scale multimodal driving benchmarks such as KITTI, nuScenes, and Waymo Open Dataset predominantly capture clear-weather conditions, limiting their use for evaluating perception under real distribution shift from adverse weather. Two existing datasets focus specifically on snowy conditions: 180 the Canadian Adverse Driving Conditions dataset (CADC) [1], the first fully snowy dataset built for 3D object detection, and the Winter Adverse Driving dataSet (WADS), the first to add 3D semantic segmentation and individual-snowflake labeling; both, however, are reported to have comparatively low-resolution or unimodal sensing relative to more recent captures. Bijelic et al.’s 185 fog- and rain-focused multimodal fusion work [13] addresses an adjacent adverse-weather regime with a different sensing challenge (fog/rain rather than snow) and is not built around a clear-to-degraded onset transition. Snowy Scenes [2], described fully in Section 4, is a substantially more recent (2026) multimodal dataset captured entirely in real snowy conditions with 128-beam LiDAR, RGB, 190 and thermal cameras; the price of that specificity, as we found once real access arrived, is that it contains no clear-weather split at all, which this paper addresses by measuring a real, physically grounded severity ordering across its own weather categories rather than assuming the dataset would hand us the clear/snow contrast originally planned around. This paper’s own contribution 195 to that evaluation landscape is not a new dataset. We apply the null-detector diagnostic to both Snowy Scenes and CADC: on Snowy Scenes it detects a genuine content artifact that the real detector does not show, and on CADC’s corrected, drive-disjoint evaluation neither the real nor the null detector reaches significance, so the diagnostic does not fire spuriously when the honest answer is that 200 we cannot tell. The monitoring measurements themselves rest on CADC alone, and Snowy Scenes serves as a negative control rather than a second positive result.

### 3. Method

#### 3.1. Background: The Monitored Detector

205 The detector this paper monitors is a camera/LiDAR BEV fusion architecture; we summarize only the components this paper’s monitoring layer directly depends on, since the monitor itself is detector-agnostic and this paper’s own contribution begins with the monitoring layer described in the rest of this section.

210 Camera and LiDAR streams are each encoded into a shared bird’s-eye-view (BEV) feature grid,  $F_{\text{cam}}, F_{\text{lid}} \in \mathbb{R}^{H_b \times W_b \times C}$ . A Cross-modal Consistency Probe (CCP) computes, per BEV grid cell  $(i, j)$ , a consistency score

$$S(i, j) = \sigma \left( \text{MLP}_S [F_{\text{cam}}(i, j) \oplus F_{\text{lid}}(i, j) \oplus |F_{\text{cam}}(i, j) - F_{\text{lid}}(i, j)|] \right) \in [0, 1], \quad (1)$$

from the concatenated ( $\oplus$ ) camera and LiDAR feature vectors at that cell and their absolute difference, so that  $1 - S(i, j)$  is a cross-modal disagreement signal intended to flag localized geometric or semantic mismatch between the two modalities. A Gated Adaptive Fusion Module (GAFM) uses  $S$  to weight each modality’s contribution to the fused BEV representation at the grid-cell level, and an Adversarial Consistency Training (ACT) objective trains the detector, the CCP, and the fusion gate jointly under a loss combining detection accuracy, a 215 CCP contrastive-consistency term, and an attribution-regularization term, with adversarially perturbed camera and LiDAR branches included in the training loop to encourage  $S$  to remain informative under perturbation. This paper’s own contribution begins with the next subsection: everything from here on – the conformal calibration, the sequential test, the spatial multiplicity control, 220 and the use (and, as Section 4 reports, the eventual disuse) of  $S$  as a monitoring covariate – is this paper’s method, evaluated against real data in Section 4.

Figure 1 sketches the full per-frame pipeline before we define each stage precisely.

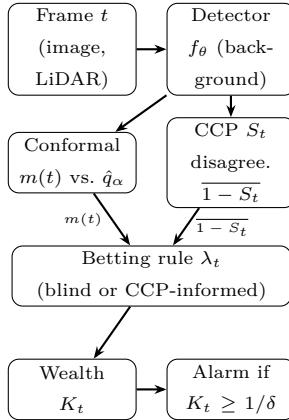


Figure 1: Per-frame monitoring pipeline. The detector  $f_\theta$  (dashed conceptually from the rest, since it is background infrastructure – Section 3.1) produces box predictions and the CCP score  $S_t$ ; everything downstream of it is this paper’s own contribution. The spatial variant (Section 3.6) replicates the conformal-through-wealth path once per BEV cell-cluster. The betting rule is either covariate-blind (uses only the  $m(t)$  path) or CCP-informed (also uses the disagreement path).

### 3.2. Problem Setup

230 We observe a stream of driving frames indexed by  $t = 1, 2, \dots$ , one per LiDAR sweep. At every  $t$ , the base 3D detector  $f_\theta$  described above produces per-object box predictions from the frame’s fused camera/LiDAR BEV features, via a center-point / heatmap detection head in the style of [14, 15]: a per-cell object-presence heatmap  $H$  locates candidate object centers and a co-located regression map  $B$  predicts each object’s box parameters at that cell; the monitor itself is detector-agnostic and does not depend on  $f_\theta$ ’s specific architecture beyond this head structure, which is what lets Section 4.2’s null-detector control zero  $B$  while leaving  $H$  intact. We wrap those predictions in split conformal prediction, calibrated once on a held-out set of frames drawn from the nominal (least-degraded) regime available in the data. For object  $i$  in frame  $t$ , the nonconformity score is the  $\ell_1$  box-regression residual restricted to grid cells a ground-truth object actually occupies,

$$s_i(t) = \|\hat{B}_i(t) - B_i(t)\|_1,$$

and the calibrated threshold  $\hat{q}_\alpha$  is the usual finite-sample-valid split-conformal quantile: the  $\lceil (n+1)(1-\alpha) \rceil / n$  empirical quantile of the calibration set's non-  
245 conformity scores  $\{s_i\}_{i=1}^n$ , so that a new object is covered iff  $s_i(t) \leq \hat{q}_\alpha$ . The frame-level miscoverage rate is

$$m(t) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}\{s_i(t) > \hat{q}_\alpha\},$$

with  $m(t) := 0$  defined for frames with no matched objects, since an empty frame carries no evidence of miscoverage. The null hypothesis the monitor tests is that the pipeline is still operating nominally,  $H_0 : \mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$  for all  
250  $t$ ; the goal is to detect the first violation – weather-onset degradation – with an anytime-valid false-alarm guarantee, not merely a guarantee at one fixed check time.

### 3.3. The $e$ -Process and Ville's Inequality

The wealth process is

$$K_0 = 1, \\ K_t = \prod_{s=1}^t (1 + \lambda_s(m(s) - \alpha)), \quad \lambda_s \in [0, \tfrac{1}{\alpha}] \text{ chosen } \mathcal{F}_{s-1}\text{-measurably.} \quad (2)$$

255 We work through why this construction gives an anytime-valid test, rather than only stating the result, since the two steps – the supermartingale property and Ville's inequality – are each short but easy to state imprecisely.

*Step 1:  $K_t$  is a nonnegative supermartingale under  $H_0$ .* Non-negativity requires  $1 + \lambda_t(m(t) - \alpha) \geq 0$  for every realizable  $m(t) \in [0, 1]$ . Since  $\lambda_t \geq$   
260 0, the expression is affine and non-decreasing in  $m(t)$ , so its minimum over  $m(t) \in [0, 1]$  occurs at  $m(t) = 0$ , giving  $1 - \lambda_t \alpha$ . This is non-negative exactly when  $\lambda_t \leq 1/\alpha$  – the bound our implementation enforces on every betting rule (`WealthProcess.lambda_max` in `conformal_monitor/betting.py`), and the reason the bound is  $1/\alpha$  specifically, not the also-plausible-looking  $1/(1-\alpha)$   
265 (which only bounds the opposite endpoint,  $m(t) = 1$ , and is too loose at the

endpoint that actually binds). For the supermartingale property itself,  $\lambda_t$  is chosen  $\mathcal{F}_{t-1}$ -measurably – it may depend on the entire history up to and including frame  $t - 1$ , but not on  $m(t)$  itself – so, conditioning on  $\mathcal{F}_{t-1}$ ,

$$\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] = K_{t-1} \cdot \mathbb{E}[1 + \lambda_t(m(t) - \alpha) \mid \mathcal{F}_{t-1}] = K_{t-1}(1 + \lambda_t(\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] - \alpha)),$$

using that  $\lambda_t$  and  $K_{t-1}$  are both  $\mathcal{F}_{t-1}$ -measurable and can be pulled outside the conditional expectation. Under  $H_0$ ,  $\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$ , so the bracketed term is  $\leq 1$  whenever  $\lambda_t \geq 0$  (which it always is, by construction), giving  $\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] \leq K_{t-1}$ : precisely the supermartingale inequality, holding for *every* choice of predictable, non-negative  $\lambda_t \leq 1/\alpha$ , which is exactly the freedom the different betting rules in Sections 3.4–3.5 exploit without ever needing to reprove this step.

*Step 2: Ville's inequality bounds the running maximum..* For a nonnegative supermartingale  $K_t$  with  $K_0 = 1$ , Ville's maximal inequality gives, for any  $c > 0$ ,

$$\mathbb{P}_{H_0}\left(\sup_{t \geq 0} K_t \geq c\right) \leq \frac{\mathbb{E}[K_0]}{c} = \frac{1}{c}.$$

Setting  $c = 1/\delta$  gives  $\mathbb{P}_{H_0}(\sup_t K_t \geq 1/\delta) \leq \delta$ . The alarm rule – reject  $H_0$  the first time  $K_t \geq 1/\delta$  – therefore has false-alarm probability at most  $\delta$  *regardless of the (random) stopping time at which the process is examined*: the guarantee is over the entire trajectory's supremum, not over  $K_t$  at one pre-specified  $t$ , which is exactly what makes checking continuously, with no fixed monitoring horizon and no correction for how many times the test has been evaluated, statistically valid rather than an uncorrected multiple-testing error. This two-step argument – the supermartingale property from Step 1, Ville's inequality from Step 2 – is the established backbone of testing by betting and is not this paper's contribution; what varies across the rest of this section, and what this paper's own contribution actually depends on, is how the predictable sequence  $\lambda_t$  is chosen.

### 3.3.1. A Worked Numeric Example

To build intuition before the general betting rules, we trace  $K_t$  over five synthetic frames by hand, with  $\alpha = 0.2$  (so  $\lambda_{\max} = 5$ ) and a fixed, non-adaptive  $\lambda_t = 2$  for simplicity (an actual bettor updates  $\lambda_t$  per Sections 3.4–3.5; holding it fixed here isolates what the recursion itself does). Suppose the observed  
 295 miscoverage sequence is  $m = (0.1, 0.15, 0.6, 0.55, 0.5)$  – nominal for the first two frames, then abruptly degraded from frame 3 onward, mimicking a weather onset at  $t = 3$ :

$$\begin{aligned} K_1 &= K_0(1 + 2(0.10 - 0.2)) = 1 \times 0.80 = 0.80, \\ K_2 &= K_1(1 + 2(0.15 - 0.2)) = 0.80 \times 0.90 = 0.72, \\ K_3 &= K_2(1 + 2(0.60 - 0.2)) = 0.72 \times 1.80 = 1.296, \\ K_4 &= K_3(1 + 2(0.55 - 0.2)) = 1.296 \times 1.70 = 2.203, \\ K_5 &= K_4(1 + 2(0.50 - 0.2)) = 2.203 \times 1.60 = 3.525. \end{aligned}$$

Two things are worth noticing in this trace. First, wealth *shrinks* while  $m(t) < \alpha$   
 300 (frames 1–2): a bettor staking  $\lambda_t > 0$  on “miscoverage will exceed  $\alpha$ ” loses money, by construction, when the pipeline is behaving nominally – this is the price of testing at all, and is what keeps the process a supermartingale rather than an unconditionally growing one. Second, wealth grows multiplicatively, not additively, once  $m(t) > \alpha$  consistently (frames 3–5): each factor  $> 1$  compounds  
 305 against the previous wealth, which is why a real onset that persists for several frames can cross a large threshold  $1/\delta$  (e.g.  $1/\delta = 20$  for  $\delta = 0.05$ ) well before  $t \rightarrow \infty$ , even though no single frame’s factor is large by itself. In this toy trace,  $K_5 = 3.525$  would already trigger an alarm at  $\delta = 0.3$  ( $1/\delta \approx 3.33$ ) but not yet at  $\delta = 0.1$  or  $\delta = 0.05$  – illustrating concretely why detection delay lengthens as  
 310  $\delta$  shrinks: a smaller false-alarm budget requires more accumulated evidence (a higher wealth threshold) before the anytime-valid guarantee permits an alarm.

### 3.4. Covariate-Blind Baselines

Two covariate-blind betting rules choose  $\lambda_t$  purely from the history of  $m(t)$  itself, reproducing [8] as the baseline to beat. Approximate GRAPA, with  $X_t =$

315  $m(t) - \alpha$ , sets

$$\lambda_t = \text{clip}\left(\frac{\bar{X}_{<t}}{\mathbb{E}[X_{<t}^2] + \varepsilon}, 0, \lambda_{\max}\right),$$

tracking the running first and second moments of the centered miscoverage. Scale-Free Online Gradient Descent instead performs AdaGrad-style updates on the log-wealth gradient, with  $g_t = X_t/(1 + \lambda_t X_t)$  and  $\eta_t = \lambda_{\max}/\sqrt{1 + \sum_{s \leq t} g_s^2}$ ,

$$\lambda_{t+1} = \text{clip}(\lambda_t + \eta_t g_t, 0, \lambda_{\max}).$$

Both are exact reimplementations of the adaptive betting rules [8] uses, run  
320 here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on.

### 3.5. Covariate-Informed Betting

Alongside the two contributions in Sections 3.6 and 4, we also investigated conditioning the betting rate on an external signal available at the same timestep  
325 as  $m(t)$  but computed from independent evidence: the disagreement output of the monitored detector’s own CCP,  $1 - S$  (Equation 1). We describe the mechanism because it motivates this paper’s monitoring design, and note here that it is specified and implemented but not evaluated in this paper (Section 4.7). Our covariate-informed bettor wraps either covariate-blind base bettor and scales its  
330 stake by the frame’s mean disagreement,

$$\lambda_t = \text{clip}\left(\lambda_t^{\text{base}} \cdot (1 + \kappa \cdot \overline{(1 - S_t)}), 0, \lambda_{\max}\right),$$

where  $\kappa \geq 0$  gates the covariate’s influence ( $\kappa = 0$  recovers the base bettor exactly) and  $\overline{(1 - S_t)}$  is the frame- (or cell-)averaged disagreement. The intended mechanism is that a scene beginning to degrade shows rising cross-modal disagreement before enough individual object detections have actually missed coverage for  $m(t)$  to move – disagreement as a leading indicator, miscoverage as  
335 the lagging one the covariate-blind rules are stuck reacting to. Whether that intended mechanism holds on real data is not settled in this paper; the rule is specified and implemented, and no result for it is reported here (Section 4.7).

Figure 2 shows why we did not lean on this covariate, on real frames rather  
 340 than in the abstract. It shows six real Snowy Scenes val-split frames run through  
 the trained checkpoint — three **accumulated** (nominal) and three **falling** (de-  
 graded), spread across each category’s frame pool rather than consecutive sam-  
 ples — with the camera image, the LiDAR bird’s-eye-view occupancy channel,  
 and the CCP disagreement map  $1 - S$  for each. The per-frame mean disagree-  
 345 ment is 0.611, 0.583, 0.576 for the three **accumulated** frames and 0.564, 0.559,  
 0.573 for the three **falling** frames — category means of 0.590 (nominal) ver-  
 sus 0.565 (degraded), consistently the opposite of the ordering the covariate is  
 meant to provide, and consistent across all six frames rather than an artifact  
 of one pair. Our best explanation is that the CCP is trained to stay invariant  
 350 under an adversarial-robustness objective (Section 3.1), which is a qualitatively  
 different feature-space disturbance from real snowfall, and invariance learned  
 against one does not obviously transfer to sensitivity toward the other. This  
 is why Section 4.7 declines to report a result for the covariate-informed bettor  
 rather than reporting one anyway.

### 355 3.6. Spatially Resolved Monitoring with Online Multiplicity Control

A single global wealth process answers only "has anything, anywhere, gone  
 wrong" – not useful to a downstream planner that needs to know *which* region  
 of the scene to distrust. We instead partition the BEV grid into an  $(n_h \times n_w)$   
 layout of cell-clusters (average-pooling per-cell miscoverage and disagreement  
 360 down to this resolution) and run one independent sequential tester per cluster,  
 each with its own wealth process and its own instance of whichever bettor is  
 under test. Testing  $n = n_h n_w$  hypotheses simultaneously, every frame, requires  
 multiplicity control or the overall false-alarm rate inflates with  $n$ ; we support  
 two regimes. Bonferroni allocates each cell a fixed budget  $\delta/n$  and, once a cell  
 365 alarms, marks it permanently flagged – valid under arbitrary dependence but  
 conservative, and the sticky-once-alarmed semantics is what we use for the per-  
 cell detection-delay evaluation regardless of which regime produced the alarm.  
 The e-BH alternative treats each cell’s current wealth  $K_t^{(i)}$  as an e-value: for



a nonnegative supermartingale started at  $K_0 = 1$ ,  $\mathbb{E}_{H_0}[K_t^{(i)}] \leq \mathbb{E}_{H_0}[K_0^{(i)}] = 1$   
 370 at any fixed  $t$  (immediate from Section 3.3's Step 1, since the supermartingale  
 inequality applied recursively from  $t = 0$  gives  $\mathbb{E}[K_t] \leq \mathbb{E}[K_{t-1}] \leq \dots \leq K_0 = 1$ )  
 – exactly the definition of an e-value under  $H_0^{(i)}$ : a nonnegative random variable  
 with expectation at most 1. Given  $n = n_h n_w$  such e-values at a fixed  $t$ , one per  
 cell, e-BH [16] rejects the set

$$R_t = \left\{ i : K_t^{(i)} \geq \frac{n}{k^* q} \right\}, \quad k^* = \max \left\{ k \in \{0, \dots, n\} : e_{(k)} \geq \frac{n}{k q} \right\},$$

375 where  $e_{(1)} \geq e_{(2)} \geq \dots \geq e_{(n)}$  is the descending order statistic of the  $n$  cur-  
 rent wealths, and  $k^* = 0$  gives  $R_t = \emptyset$ . The rule is a direct e-value analogue  
 of Benjamini-Hochberg: it thresholds the  $k$ -th largest e-value against a level  
 that grows with  $k$  (more permissively for larger candidate rejection sets), then  
 commits to the *largest* such  $k$  – the same "largest crossing index" structure as  
 380 classical BH's  $p$ -value threshold, but with the inequality direction reversed be-  
 cause e-values are large, not small, under evidence against the null. Its false  
 discovery rate guarantee,  $\mathbb{E}[\text{FDR}_t] \leq q$  at every fixed  $t$ , holds under *arbitrary*  
*dependence* among the  $n$  e-values – unlike classical BH on  $p$ -values, which re-  
 quires independence or specific positive-dependence structures (PRDS) to con-  
 385 trol FDR without an extra  $\log n$  correction factor – because the e-value defi-  
 nition ( $\mathbb{E}[e_i] \leq 1$  under  $H_0^{(i)}$ ) is closed under averaging regardless of the joint  
 dependence structure of the  $e_i$ , which is precisely the guarantee needed here:  
 spatially adjacent BEV cells' miscoverage and disagreement signals are driven  
 by overlapping physical causes (a single snow squall affects many neighboring  
 390 cells at once) and are certainly not independent. Unlike Bonferroni's sticky  
 once-alarmed flags, e-BH is applied fresh at every timestep from each cell's cur-  
 rent wealth, so it produces a live, non-sticky map that can un-flag a cell once its  
 local evidence subsides – appropriate for a spatial map meant to reflect *current*,  
 not historical, trustworthiness. Algorithm 1 summarizes the per-frame update.

### 395 3.6.1. Betting-Rule Comparison: Theoretical Properties and Tradeoffs

Table 1 summarizes the three betting rules evaluated in this paper, all valid choices of the predictable  $\lambda_t$  in Equation 2 (so all three inherit Section 3.3’s anytime-valid guarantee unconditionally; what differs is statistical power, not validity). AGRAPA is a plug-in estimator of the (asymptotically) growth-  
400 rate-optimal constant bet, tracking the running first and second moments of  $X_t = m(t) - \alpha$  and converging quickly when the true miscoverage rate is close to stationary, but reacting relatively slowly to an abrupt regime change, since a single new observation only nudges a running average computed over all prior history. SF-OGD instead takes an online-convex-optimization view, performing  
405 AdaGrad-style gradient ascent directly on (an approximation to) the log-wealth objective with a decaying step size; this makes it more reactive to recent observations by construction (the step size  $\eta_t$  shrinks as  $1/\sqrt{t}$ , but each step still moves  $\lambda_t$  in the direction the *most recent* gradient indicates), at the cost of higher variance in  $\lambda_t$  itself, particularly early in a stream before  $\sum_{s \leq t} g_s^2$  has accumulated  
410 enough to stabilize  $\eta_t$ . Both are exact reimplementations of the rules [8] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on. The CCP-informed variant wraps either base rule multiplicatively rather than replacing it, so it inherits whichever base rule’s reactivity/stability tradeoff was chosen, and adds a single  
415 scalar gain  $\kappa$  controlling how strongly the covariate perturbs the base bet;  $\kappa = 0$  recovers the base rule exactly, which would make a covariate-informed versus covariate-blind comparison a controlled one (same base rule, same wealth-process mechanics, differing only in whether  $\overline{1 - S_t}$  enters the stake) rather than two unrelated procedures, were such a comparison reported.

### 420 3.7. Session-Conditional (Mondrian) Calibration

Split conformal calibration pools nonconformity scores and produces one quantile, valid on average over the calibration distribution. Real driving data arrives in sessions that differ persistently in weather, lighting, traffic and route, and Section 4.3 measures the resulting per-session violation directly. Mondrian

Table 1: Betting rules compared in this paper. All three are valid choices of  $\lambda_t$  in Equation 2 and inherit the same anytime-valid guarantee unconditionally (Section 3.3); they differ in statistical power, not validity.

| Rule         | Mechanism                                                  | Reactivity / stability                                                                                     |
|--------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| AGRAPA       | running 1st/2nd moment plug-in                             | smooth, slower to react to abrupt shift                                                                    |
| SF-OGD       | AdaGrad-style gradient ascent on log-wealth                | more reactive, higher early-stream variance                                                                |
| CCP-informed | multiplicative covariate gain $\kappa$ on either base rule | inherits base rule’s trade-off; adds covariate sensitivity (specified and implemented; not evaluated here) |

425 conformal prediction [12, 3] conditions calibration on a taxonomy, giving validity within each category; here the category is the driving session.

Mondrian requires calibration data in every category it is queried about, which forces a within-session split: each session calibrates on its own leading frames and is monitored on the remainder. This is the deployment-faithful  
430 arrangement — a vehicle calibrates on the early known-nominal portion of the drive it is currently on, rather than assuming a quantile estimated on previous drives transfers — and degraded frames are scored against the quantile of the session the stream began in, which is what a deployment does when conditions change mid-drive. Because calibration and monitored frames then sit adjacent  
435 in time within a correlated stream, we insert a temporal gap between them and sweep its size; an effect inflated by adjacency shrinks as the gap grows.

### 3.8. A Phantom-Detection Nonconformity Score

The score of Section 3.2 is a box-regression residual restricted to cells a ground-truth object occupies, and is therefore structurally blind to spurious  
440 detections in empty space. Snow’s dominant effect on LiDAR is precisely spurious near-range returns, so we add a second score family: the predicted peak

---

**Algorithm 1** Per-frame spatial monitoring update

---

**Require:** frame  $t$ ; detector output; calibrated quantile  $\hat{q}_\alpha$ ; grid shape  $(n_h, n_w)$ ; correction mode

- 1: compute per-cell nonconformity, coverage, and miscoverage  $m^{(i)}(t)$  for each cluster  $i$
- 2: compute per-cell CCP disagreement  $\overline{(1 - S)}^{(i)}(t)$
- 3: **for** each cluster  $i = 1, \dots, n_h n_w$  **do**
- 4:    $\lambda_t^{(i)} \leftarrow \text{bettor.NEXT\_LAMBDA}(\overline{(1 - S)}^{(i)}(t))$
- 5:    $K_t^{(i)} \leftarrow K_{t-1}^{(i)} \cdot (1 + \lambda_t^{(i)}(m^{(i)}(t) - \alpha))$
- 6:   bettor.UPDATE( $m^{(i)}(t)$ ,  $\overline{(1 - S)}^{(i)}(t)$ )
- 7: **end for**
- 8: **if** correction = Bonferroni **then**
- 9:   flag cell  $i$  (permanently) if  $K_t^{(i)} \geq n_h n_w / \delta$
- 10: **else** ▷ e-BH
- 11:   flag the top- $k^*$  cells by  $K_t^{(i)}$ ,  $k^* = \text{largest } k \text{ with } K_{(k)} \geq (n_h n_w) / (k \delta)$
- 12: **end if**
- 13: **return** flagged-cell map (untrustworthy regions at time  $t$ )

---

class activation at cells containing no ground-truth object, calibrated by the same split-conformal procedure at level  $\alpha$ . The frame-level rate is the convex combination

$$m(t) = (1 - w) m_{\text{loc}}(t) + w m_{\text{phantom}}(t), \quad (3)$$

445 with  $w \in [0, 1]$ . Each component is calibrated at level  $\alpha$  and so has expectation at most  $\alpha$  under the nominal distribution; a convex combination of two such quantities inherits the bound, which is exactly the  $H_0$  condition the wealth process of Section 3.3 requires. No new theory is needed: the conformal guarantee is agnostic to the choice of score function. Setting  $w = 0$  recovers the original  
450 score exactly.

### 3.9. Block Aggregation

Averaging  $m$  over blocks of  $k$  consecutive frames before the betting step trades temporal resolution for reduced variance in the quantity the bettor consumes. Under independent frames the reduction would be  $\sqrt{k}$ ; driving frames  
455 are strongly correlated, so the realised gain is smaller; we do not report a measurement of it here. We keep one trajectory entry per frame regardless of  $k$ , so

detection delays remain in frame units and are comparable across block sizes and against the unblocked baseline.

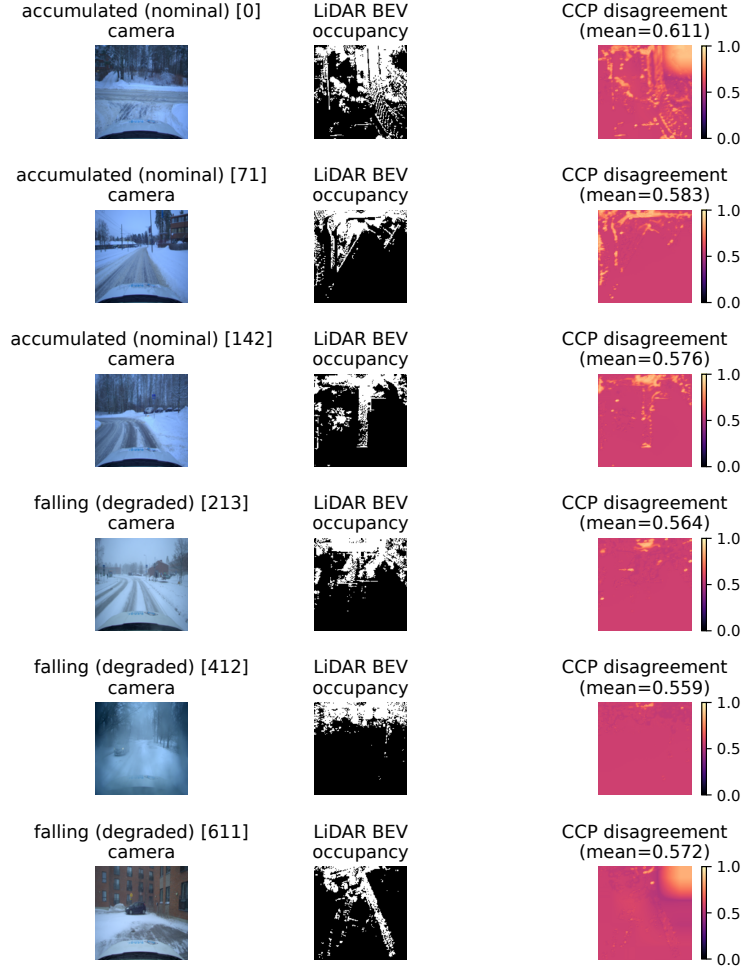


Figure 2: Six real Snowy Scenes val-split frames run through the trained checkpoint: three **accumulated** (nominal) and three **falling** (degraded), each showing the camera image, LiDAR BEV occupancy channel, and the CCP disagreement map  $1 - S$ . Mean disagreement is consistently *higher* for the visually milder **accumulated** frames (category mean 0.590) than the visually more severe **falling** frames (category mean 0.565) — the wrong-direction pattern this section diagnoses, visible here across six independent real frames rather than only in aggregate.

## 4. Experiments

460 Every number below is produced by a script committed alongside the manuscript  
and written to a JSON artifact named in the corresponding caption.

### 4.1. Setup

CADC [1] provides 32 downloaded driving sessions across three collection  
dates: 18 nominal (**bare-road**) and 14 snow-covered. Snowy Scenes [2] pro-  
465 vides real snowy driving; it contains no clear-weather split, so its contrast is  
constructed across weather categories and it serves here as a negative control  
rather than a second positive result.

*Train/evaluation separation..* The detector and the monitor are built on dis-  
joint *drives*. Frame-level holdout would not suffice: consecutive CADC frames  
470 are  $\sim 10$  Hz samples of one scene, so a frame-level split leaves near-duplicates  
of training frames in evaluation. Drives are interleaved within each weather  
category, giving 16 training and 16 evaluation drives (1534 / 1408 frames) with  
both categories and all three dates represented on each side. Calibration and  
monitoring both draw only from evaluation drives, so the conformal scores are  
475 out-of-sample.

This matters because an earlier version of this work did not do it. The detec-  
tor and the monitor were constructed over one unsplit dataset object, so every  
calibration, nominal and degraded frame had been a training frame, and the  
nonconformity scores were in-sample residuals against which a calibrated quan-  
480 tile bounds nothing. A regression test now asserts that the train and evaluation  
frame-id sets are disjoint.

*Nesting with session-conditional calibration..* Mondrian calibration needs cal-  
ibration frames from inside each monitored session, so an evaluation drive is  
split again internally into a leading calibration prefix, a temporal gap, and a  
485 monitored remainder. That inner split lives entirely within evaluation drives  
and never reaches training data.

Table 2: Null-detector diagnostic. The nominal-to-degraded jump in  $m$  is identical to 17 significant figures whether the box head’s real output or zeros are used, on both datasets — the monitor’s evidence does not depend on the detector at all. Artifacts: `detector_collapse_cadc.json`, `detector_collapse_snowy.json`.

| Dataset      | real detector        | zeroed detector      | difference |
|--------------|----------------------|----------------------|------------|
| CADC         | 0.13904668284371138  | 0.13904668284371138  | 0          |
| Snowy Scenes | −0.15573322510822504 | −0.15573322510822504 | 0          |

*Inference.* Sessions, not frames, are the unit of analysis: frames within a drive are strongly correlated and frame-level resampling understates intervals badly. Where an estimate contrasts nominal against degraded data, *both* arms are re-sampled — nominal sessions and degraded drives — because the degraded side carries real between-drive variation (measured sd 0.128, Mondrian calibration) that a fixed pool would silently treat as zero. All intervals are percentile bootstrap.

#### 4.2. The Null-Detector Diagnostic

The diagnostic reruns the entire pipeline, calibration included, with the detector’s box-regression output replaced by zeros. The nonconformity score then reduces to  $\|B_{\text{target}}\|_1$  and depends only on ground-truth scene content. If a monitor’s behaviour reflects detector error, zeroing the detector must change it.

Table 2 gives the result on the checkpoints that produced our earlier numbers. The agreement is exact, not approximate: the detector contributes nothing measurable to the monitor’s evidence, because those checkpoints had collapsed under a placeholder detection loss and *were* null detectors. This is the cleanest available statement of the finding, and it is immune to any question about evaluation-harness configuration, since it compares two branches of a single deterministic computation.

We draw a conclusion beyond our own case. A conformal monitor built on a degenerate score does not fail conspicuously — it reports calibrated quantiles and plausible alarm times throughout. One extra evaluation pass distinguishes



Table 3: Per-drive mean frame miscoverage under one pooled quantile ( $\hat{q} = 2.712$ , target  $\alpha = 0.20$ ), on the drive-disjoint evaluation split (the detector’s training drives never appear here). All **bare** drives are nominal clear-road driving. Artifacts: `per_drive_miscoverage_cadc.json`, `table_summary_cadc.json`.

| Group              | $n$ drives | mean $m$ (sd) | range         |
|--------------------|------------|---------------|---------------|
| calibration drives | 5          | 0.253 (0.180) | 0.113 – 0.522 |
| nominal drives     | 4          | 0.217 (0.118) | 0.114 – 0.386 |
| degraded drives    | 7          | 0.299 (0.157) | 0.107 – 0.479 |

the two situations unambiguously. We suggest it be reported as standard, with  
510 one scope condition stated honestly: this is a null for scores built from regression  
residuals. A score reading a different output head needs its own null, and  
Section 5.4 explains why constructing one is not always straightforward.

#### 4.3. Cross-Session Exchangeability Failure

An earlier version of this table was computed on a checkpoint trained on all  
515 32 drives, including every drive shown here; every number below is recomputed  
on the checkpoint that never saw them, and the picture that survives is smaller  
than what we first reported. With only 9 held-out **bare** drives split 5/4 between  
calibration and the nominal stream, the nominal group is thin — we say so  
plainly, because a 3–4 $\times$  spread over four drives is not the kind of claim that  
520 should be dressed up as more than it is. One of four nominal drives sits above  
 $\alpha$ ; the spread across all held-out **bare** drives (calibration and nominal together)  
is 4.6 $\times$ , and restricted to the nominal drives alone, 3.4 $\times$ . The comparison  
that matters is against the effect being sought: the degraded-minus-nominal  
difference is +0.082 while the nominal-drive standard deviation is 0.118, and all  
525 four nominal drives fall inside the degraded drives’ range. Even on this smaller,  
correctly held-out sample, a monitor calibrated marginally across sessions is  
detecting session identity to a degree comparable with the effect it exists to  
find, and its alarms carry route information rather than perception information  
— the qualitative conclusion is unchanged even though the quantitative spread

Table 4: Mondrian versus marginal calibration on the drive-disjoint checkpoint, matched on sessions (9), frames, gap and cached scores, differing only in which quantile scores each session.

Artifact: `mondrian_gap_sweep_cadc.json`.

|     | effect / nominal sd |          | worst $ m - \alpha $ |              |
|-----|---------------------|----------|----------------------|--------------|
| gap | marginal            | Mondrian | marginal             | Mondrian     |
| 0   | <b>0.980</b>        | 0.781    | 0.259                | <b>0.230</b> |
| 10  | <b>1.020</b>        | 0.620    | <b>0.206</b>         | 0.279        |
| 20  | <b>0.864</b>        | 0.474    | <b>0.231</b>         | 0.383        |

530 is smaller than we first reported.

#### 4.4. Session-Conditional Calibration: An Inconsistent Fix

Mondrian conformal prediction [12] conditions the quantile on a taxonomy; here the taxonomy is the driving session. It is the natural response to Section 4.3’s finding, so we evaluate it on the same drive-disjoint checkpoint, 535 matched session for session, frame for frame and gap for gap against marginal calibration.

The result does not support a clean recommendation either way, and we report it as such rather than picking the reading that flatters the method. On worst-case per-session deviation from  $\alpha$  — the statistic Mondrian calibration 540 is designed to control — it wins narrowly at gap 0 and loses at gaps 10 and 20, the opposite of what conditioning on session identity should buy as the gap widens and the twenty calibration frames per session have less contamination from adjacency to explain a benefit. On the ratio that matters for detecting the weather effect itself, effect over nominal standard deviation, Mondrian is 545 *worse* than marginal at every gap, because splitting twenty calibration frames per session into its own quantile estimate adds quantile-estimation noise that a single pooled quantile does not pay. Session-conditional calibration is not a free lunch: with this few calibration frames per session, it can trade one source of miscalibration (between-session heterogeneity) for another (per-session quantile-

Table 5: Nominal-versus-degraded effect on drive-disjoint held-out drives (9 nominal sessions, 7 degraded drives), Mondrian calibration. The two-level interval resamples nominal sessions and degraded drives; the nominal-only interval resamples sessions alone, from the same draws, and is shown so the effect of the correction is visible. Artifact: `split_effect_split_ep5_fixed.json`.

| gap | effect | two-level CI       | nominal-only CI    | inflation    | null det. |
|-----|--------|--------------------|--------------------|--------------|-----------|
| 0   | +0.069 | $[-0.073, +0.210]$ | $[-0.039, +0.163]$ | $1.40\times$ | n.s.      |
| 10  | +0.058 | $[-0.090, +0.203]$ | $[-0.060, +0.160]$ | $1.33\times$ | n.s.      |
| 20  | +0.045 | $[-0.106, +0.197]$ | $[-0.074, +0.157]$ | $1.31\times$ | n.s.      |

estimation noise), and which one dominates is not obvious in advance. Neither  
statistic carries an interval —  $\max_s |m_s - \alpha|$  and the effect-to-noise ratio are  
both computed from nine sessions, among the least stable estimators available  
at this sample size — so a practitioner should measure this on their own model  
and calibration budget rather than assume either direction from a single run.

#### 4.5. The Weather Effect: an Underpowered Null

The effect is not distinguishable from zero at any gap, under either calibration scheme, and the null detector is likewise null throughout.

*This is an underpowered null, not a demonstrated absence..* The interval  
 $[-0.073, +0.210]$  is consistent with a real effect of the size we previously claimed  
and equally consistent with none. We compute the minimum detectable effect  
directly from this interval’s bootstrap standard error (half-width divided by  
 $z_{0.975}$ , then scaled by  $z_{0.975} + z_{0.80} \approx 2.80$  for 80% power) rather than asserting it: at gap 0,  $\text{MDE}_{80} \approx 0.202$  — almost  $3\times$  the point estimate of +0.069  
(`mde_cadc.json`). The design cannot resolve an effect of this magnitude, and  
no conclusion about the presence or absence of weather-induced degradation  
should be drawn from it.

*Why an earlier version appeared significant..* Two independent defects were present before correction, and we can only cleanly separate one

of them. The train/evaluation overlap put the entire pipeline in-  
570 sample over 18 nominal-regime sessions, giving  $+0.074$ ,  $[+0.020, +0.117]$   
(`mondrian_gap_sweep_cadc_presplit_insample.json`, preserved separately  
from the corrected artifact of the same name). A separate bookkeeping bug  
in the two-level bootstrap let each session’s first twenty frames — its own cali-  
bration set — re-enter the monitored window at gap 0, biasing that specific cell;  
575 holding the drive-disjoint split fixed and only fixing this bug moves the gap-0  
estimate from  $+0.064$  (`split_effect_split_ep5.json`, preserved for compar-  
ison) to  $+0.069$  (`split_effect_split_ep5_fixed.json`), a small change rela-  
tive to the interval width. What we cannot do cleanly is isolate the split correc-  
tion from the bug fix in combination, because the bug lived inside the two-level  
580 bootstrap that only ever ran on the held-out split; the 18-session, in-sample  
number comes from a different script entirely. What is well supported is this:  
the point estimate moves only modestly across every correction we can check  
individually ( $+0.074 \rightarrow +0.064 \rightarrow +0.069$ ), while significance is lost once the  
drive-disjoint split cuts the session count from 18 to 9 — even the nominal-only  
585 column, which does not resample the degraded arm, fails to reach significance at  
any gap on the corrected data (Table 5, fourth column). The earlier significance  
was not driven by a larger true effect; it was driven by having more sessions  
than the design was entitled to.

We suggest reporting both intervals as general practice for clustered effect  
590 estimates. A single interval tells a reader whether a result is significant; the  
pair tells them something about *why*, which is part of the difference between  
concluding “underpowered” and concluding “absent.”

#### 4.6. Snowy Scenes as Negative Control

On Snowy Scenes the real detector’s effect is  $-0.0085$  with a 95% inter-  
595 val of  $[-0.0719, +0.0553]$ , centred on zero, while the null detector’s is  $-0.1315$ ,  
 $[-0.1917, -0.0726]$  (`effect_exists_snowy.json`). The only signal present is  
a content artifact pointing the wrong way, which is why the null monitor “de-  
tected” there while the real one did not.

On the corrected CADC split (Table 5) neither arm is significant: the real  
600 detector’s interval covers zero and so does the null detector’s. This is not the  
mirror image we originally reported — an earlier draft, using the in-sample  
CADC artifact, read the real effect as significant and the null’s as covering zero,  
the reverse of Snowy. That contrast does not survive the correction, and we  
withdraw it. What the two datasets now show is asymmetric in a more modest  
605 but still useful way: Snowy Scenes demonstrates that the null-detector diagnos-  
tic *does* fire on a genuine content artifact when one is present, while CADC’s  
corrected result shows it does not fire spuriously when the honest answer is “we  
cannot tell.” Neither dataset gives us a case where the real detector is significant  
and the null is not — the comparison we would need to claim the diagnostic  
610 cleanly isolates detector-driven signal from content-driven signal in a positive  
instance — and we say so rather than reach for the reading our earlier, uncor-  
rected numbers had offered. We note that the Snowy intervals were computed  
with frame-level rather than session-level resampling and are therefore narrower  
than our convention elsewhere would give; this does not affect the qualitative  
615 reading above.

#### 4.7. What We Do Not Report

Three lines of work were carried far enough to have numbers and are ex-  
cluded, because the numbers do not support conclusions.

A curve relating measured weather effect to detector quality across check-  
620 points is omitted: its quality axis (heatmap object-versus-empty separation)  
shares a term with one of the two scores plotted against it — the phantom score  
*is* the empty-cell activation that the axis subtracts — so a decline along that  
axis is partly circular. Its inputs also predate the corrections in Section 4.1. We  
record it as an open question, not a finding.

625 A covariate-informed betting rule conditioned on the detector’s own cross-  
modal consistency score, and a spatially resolved variant using per-cell e-processes  
under multiplicity control, are described in Section 3 because they motivate the  
design, but neither is evaluated here and neither carries a result in this paper.

Operating-curve and detection-delay results are omitted because, under the  
630 corrected design, no configuration has a demonstrable false-alarm advantage:  
with 36 replicates, an observed rate of 0/36 has an exact upper confidence  
bound of 0.080, which does not establish control at  $\delta = 0.05$ .

## 5. Discussion

### 5.1. *Why a Fictitious Monitor Looks Healthy*

635 The mechanism is worth stating precisely, because it generalizes past our  
particular bug. The nonconformity score is a residual evaluated where a ground-  
truth object sits. When the prediction term vanishes it becomes  $\|B_{\text{target}}\|_1$ : a  
function of the labels alone. Nothing downstream notices. Split conformal  
calibration still returns a well-defined quantile, because it is the quantile of *some*  
640 exchangeable score distribution. The betting process still accumulates wealth,  
because the miscoverage rate still moves between the nominal and degraded  
halves of a stream. The alarm still fires at a plausible time, because ground-  
truth box statistics genuinely differ between a dataset’s splits. Every component  
behaves exactly as specified, and the monitor reports annotation structure as  
645 perception degradation.

We were careful about which dataset property drives this, having initially  
got it wrong. We had asserted object count roughly doubling between CADC’s  
nominal and degraded splits, with no measurement behind the number. Measur-  
ing it directly, on the ground-truth heatmap and on the same nominal/degraded  
650 frames used throughout (`object_counts_cadc.json`), degraded frames contain  
*fewer* matched objects than nominal ones — 12.65 versus 13.72 per frame, a ra-  
tio of 0.92, not the near-doubling we first claimed. Since the score reduces to  
 $\|B_{\text{target}}\|_1$ , object count is at most a minor contributor and the driver is more  
likely the distribution of ground-truth box dimensions and positions across the  
655 two collection dates, which we have not measured directly and report as a plu-  
sible explanation rather than a demonstrated one. It interacts directly with the  
date confound discussed below.

We do not have a clean instance of the inversion we originally reported — a real detector that alarms while a content-only null detector stays quiet, or  
 660 vice versa — once the false-alarm comparison is restricted to the corrected, drive-disjoint design: Section 4.7 explains why we do not report operating-point false-alarm rates from that design at all, and the same reasoning applies to any single number drawn from it. What we can say is narrower: scene content that is homogeneous within a category will produce a content-derived score that  
 665 looks consistent across sessions in a way genuine, session-heterogeneous detector error does not, so a null-detector control can in principle look *more* stable than the real pipeline even though it carries no information about perception at all. That is a property of what the control measures, not a claim about which arm alarmed more often on our own data — and it is still exactly why inspecting a  
 670 monitor’s outputs for smoothness cannot substitute for running the control.

## 5.2. Sessions Are Not Exchangeable, and What It Costs Navigation

Conformal validity is conditional on exchangeability, and driving data violates it in a structured way: it arrives in sessions that differ persistently in route, traffic, lighting and weather. Our measurement puts a number on the  
 675 cost — a 3–5× per-drive spread in miscoverage under a pooled quantile, on a nominal-drive sample thin enough (four held-out drives) that we do not lean on the exact multiple, with between-session variation comparable in size to the weather effect being sought.

The consequence is not a modest loss of efficiency. It is that the monitored  
 680 quantity is dominated by which session the vehicle is on, so the monitor detects session identity. For a navigation stack this is the worst available failure mode. An alarm that tracks route rather than perception fires on an unremarkable drive and stays silent on a dangerous one, and a planner wired to react — by slowing, handing back, or initiating a minimum-risk manoeuvre — inherits that  
 685 error directly and acts on it. A monitor whose sensitivity has not been demonstrated but which is nonetheless counted in a safety argument is worse than no monitor, because the assurance it appears to provide displaces mitigations that

would otherwise be required.

Two practical consequences follow for anyone deploying this pattern. First, calibration should be conditioned on the session, or at minimum validated per session rather than in aggregate — an aggregate coverage check passes while individual sessions sit far from target. Second, because our own attempt at the session-conditional fix helped on one statistic and hurt on another, and reversed direction across gaps within the *same* checkpoint (Section 4.4), the check has to be repeated against the model and calibration budget actually being shipped, not inherited from a paper — ours included.

### 5.3. Underpowered Is Not Absent

Our headline measurement is a null we decline to interpret as an absence, and the distinction carries the section. An interval of  $[-0.073, +0.210]$  around a point estimate of  $+0.069$  excludes neither a real effect of the size we previously claimed nor no effect at all. With nine evaluation sessions the minimum detectable effect at 80% power is roughly 0.20, almost  $3\times$  the point estimate, so failing to detect it is what this design does regardless of whether the effect is there.

The dual-interval reporting in Table 5 is what makes that diagnosable rather than fully assertable. Because the nominal-only and two-level intervals are computed from the same session draws and differ only in whether the degraded arm is resampled, a reader can see that even the narrower, nominal-only interval fails to exclude zero on the corrected data, and that resampling the degraded arm widens it a further  $1.3\text{--}1.4\times$  on top of that. We suggest reporting both intervals whenever an effect is estimated from clustered data on both sides; a single interval says whether a result is significant, the pair helps a reader see part of why not.

It is worth being explicit that the correction which mattered most for the point estimate was not the one we expected. Removing the train/evaluation overlap — the most serious defect, and the one that voids the earlier numbers as a matter of principle — moved the point estimate only modestly ( $+0.074 \rightarrow +0.069$  once every correction we can check is applied). Significance was lost



primarily because the drive-disjoint split leaves only nine sessions to resample, not because the true effect shrank. Both the leakage and a separate bootstrap  
720 bookkeeping bug (Section 4.5) had to be fixed on principle; neither turned out to be the dominant reason the number stopped being significant.

#### 5.4. Limitations

*Scope of the null-detector diagnostic..* It is a null for scores built from regression residuals. A score reading a different output head requires its own, and constructing one is not always straightforward: for a score built from classification-  
725 head activation, zeroing that head makes every score identical and the statistic degenerate, which is a floor rather than an informative control. We report this because it bounds the recommendation, and a general recipe — sever the dependence between the score and the model output while leaving the calibration  
730 pipeline intact — is easier to state than to instantiate for every score family.

*The diagnostic does not catch every failure..* It catches a monitor that is not reading the model. It does not catch a monitor that reads the model but measures the wrong thing — including the session-identity failure documented in Section 4.3, which passes the null-detector check.

735 *Weather is confounded with collection date..* CADC’s degraded drives come from a different date than its nominal ones, so sensor condition, route and traffic are confounded with weather. The null-detector control bounds the contribution of label content specifically; it does not bound the rest. The dataset publishes a quantitative per-drive snowfall-severity count that varies *within* date  
740 and would break this aliasing; we identified it too late to use, and regard it as the most promising next step.

*Scale and single-run training..* Nine evaluation sessions, one detector architecture, one dataset carrying the positive findings, and one training run — no seed replication, so nothing here has error bars in that dimension. Training used  
745 chunked resumption, so optimizer moment estimates re-warm at each resume, a small deviation from an uninterrupted run.

*Statistical scope.* The Snowy Scenes intervals in Section 4.6 use frame-level resampling and are narrower than our session-level convention would give. The Mondrian worst-case statistic is a maximum over nine sessions and carries no interval. We report both rather than presenting the numbers as comparable to the clustered estimates elsewhere.

## 6. Conclusion

We set out to reduce the detection delay of a conformal, anytime-valid monitor for camera/LiDAR 3D perception under real adverse weather, so that a navigation stack could be told sooner when its perception had stopped being trustworthy. We did not achieve that. What the attempt produced instead are two findings about how such monitors fail silently, and a measurement we report as unresolved rather than settled.

First, a conformal monitor can produce calibrated quantiles and plausible alarm times while measuring nothing about the detector. When the prediction head degenerates, the nonconformity score becomes a function of the ground-truth labels alone and the monitor reports annotation structure as perception degradation, with nothing in its output revealing this. The *null-detector diagnostic* — rerun the pipeline with the prediction head’s output replaced by zeros — exposes it for the cost of one evaluation pass, and on our own earlier results reproduced them bit-identically, to 17 significant figures, from a model that outputs nothing. Content that is homogeneous within a category can make a null detector’s output look no less stable than the real pipeline’s, which is precisely why output inspection cannot substitute for the control.

Second, split conformal’s exchangeability assumption fails across driving sessions. On a drive-disjoint held-out split, per-drive miscoverage under a pooled quantile spans 3–5× on nominal clear-road drives — indicative rather than precise, given only four held-out nominal drives — with between-session variation comparable to the weather effect being sought. The guarantee remains formally correct and becomes operationally useless: the monitor detects which session

the vehicle is on. For navigation that is the worst available failure mode, because a planner wired to a monitor that tracks route identity will act on route identity.

On the weather effect we report an underpowered null and are explicit that  
780 this is not an absence:  $+0.069$  with a 95% interval of  $[-0.073, +0.210]$  on drive-disjoint held-out data, consistent both with a real effect of the size we previously claimed and with none. We also show why an earlier version appeared significant, by reporting the interval with and without the degraded arm resampled — a decomposition we recommend for clustered estimates generally, because it  
785 helps distinguish an underpowered design from a genuine absence where a single interval cannot.

Three directions follow, in the order we would pursue them. The most promising is already identified: CADC publishes a quantitative per-drive snowfall-severity count that varies within collection date, which would replace the date-  
790 aliased binary contrast used here with a dose-response regression — more efficient statistically, and free of the confound that limits every effect estimate in this paper. Second, the dataset releases substantially more drives than we downloaded, and since the number of evaluation sessions is the binding constraint on every interval reported here, raising it addresses the limitation more  
795 directly than any change of estimator. Third, the null-detector diagnostic should be generalized from a single check into a family: a null per score family, and ideally a graded sensitivity profile obtained by injecting known degradations of known magnitude, so that a null result acquires a measured detection floor rather than resting on an instrument of unknown sensitivity.

800 We would emphasise one point above the others. Every failure in this work was silent — a collapsed detector, an evaluation harness whose replicates were identical, a monitor reading annotation statistics, calibration valid in aggregate and broken per session, and a train/evaluation overlap that made every conformal score in-sample. In each case the outputs looked reasonable, and in each  
805 case what exposed the problem was a control: zeroing the detector, asserting that replicates differ, measuring per session rather than in aggregate, asserting

that train and evaluation frames are disjoint. For a monitoring system, whose whole purpose is to be trusted when nothing else is watching, we think the controls deserve as much space in the reporting as the headline numbers — and  
810 that a monitor whose sensitivity has never been demonstrated should not be counted in a safety argument at all.

### **CRedit authorship contribution statement**

Author contribution roles are pending finalization of the author list.

### **Declaration of competing interest**

815 The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### **Data availability**

CADC [1] and Snowy Scenes [2] are both publicly available third-party  
820 datasets; access details are given in Section 4.1. Code implementing the null-detector diagnostic, the session-conditional calibration and two-level bootstrap procedures, and every script that produced a numbered artifact cited in this paper will be made available at a public repository upon publication.

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